Chapter 2
Determinants

2.2 Properties of Determinants

Determinants of Triangular Matrices

Theorem 1 If A is an $n \times n$ triangular matrix, then the determinant of A equals the product of the diagonal elements of A.

Expansion according to the ith row

$$|A| = \sum_{j=1}^{n} a_{ij}A_{ij} = \sum_{j=1}^{n} a_{ij}(-1)^{i+j}\det(M_{ij})$$

$$\begin{vmatrix} a_{11} & 0 & 0 & 0 \\ * & a_{22} & 0 & 0 \\ * & * & a_{33} & 0 \\ * & * & * & a_{44} \end{vmatrix}$$

- By induction:
- Check that the statement is true for 1×1 or 2×2 matrices;
- Suppose the statement holds for $(n-1)\times(n-1)$ matrices (that is, the statement is true for its minors), then apply the cofactor expansion to show that the statement is still true for $n\times n$ matrices.

Determinants of Transpose

Theorem 2 If A is an $n \times n$ matrix, then $\det(A) = \det(A^T)$.

$$|A| = \sum_{j=1}^{n} a_{ij}A_{ij} = \sum_{j=1}^{n} a_{ij}(-1)^{i+j}\det(M_{ij})$$

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \dots & a_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \dots & a_{nn} \end{vmatrix}$$

$$\begin{vmatrix} a_{11} & a_{21} & a_{31} & \dots & a_{n1} \\ a_{12} & a_{22} & a_{32} & \dots & a_{n2} \\ a_{13} & a_{23} & a_{33} & \dots & a_{n3} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & a_{3n} & \dots & a_{nn} \end{vmatrix}$$

- By induction:
- Check that the statement is true for 1×1 or 2×2 matrices;
- Suppose the statement holds for $(n-1)\times(n-1)$ matrices (that is, the statement is true for its minors), then apply the cofactor expansion to show that the statement is still true for $n\times n$ matrices.

① When $n=1$

$3^T = 3$

$|A| = |a| = a$

$|A^T| = |a^T| = |a| = a = |A|$

② When $n=2$

$$|A| = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}a_{21}$$

$$|A^T| = \begin{vmatrix} a_{11} & a_{21} \\ a_{12} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}a_{21} = |A|$$

③ Assume. when $n=k$, $|A| = |A^T|$

④ When $n=k+1$

$$|A| = \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1,k+1} \\ a_{21} & a_{22} & \cdots & a_{2,k+1} \\ \vdots & \vdots & \ddots & \vdots \\ a_{k+1,1} & a_{k+1,2} & \cdots & a_{k+1,k+1} \end{vmatrix}$$

$(k+1)\times(k+1)$

$= a_{11}A_{11} + a_{12}A_{12} + \cdots + a_{1,k+1}A_{1,k+1}$

$$|A^T| = \begin{vmatrix} a_{11} & a_{21} & \cdots & a_{k+1,1} \\ a_{12} & a_{22} & \cdots & a_{k+1,2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1,k+1} & a_{2,k+1} & \cdots & a_{k+1,k+1} \end{vmatrix}$$

$$= [A^T]_{11} A^T_{11} + [A^T]_{21} A^T_{21} + \cdots + [A^T]_{k+1,1} A^T_{k+1,1}$$

$$= a_{11} A^T_{11} + a_{12} A^T_{21} + \cdots + a_{1,k+1} A^T_{k+1,1}$$

Consider the cofactors A_{ij} and A^T_{ji}

$$A_{ij} = (-1)^{i+j} |M_{ij}|_{k \times k}$$

$$A^T_{ji} = (-1)^{j+i} |M^T_{ij}|_{k \times k} = (-1)^{i+j} |M_{ij}| = A_{ij}$$

Eg. $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$ $\quad A^T = \begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix}$

$$A_{12} = (-1)^{1+2} \begin{vmatrix} 4 & 6 \\ 7 & 9 \end{vmatrix} \qquad A^T_{21} = (-1)^{2+1} \begin{vmatrix} 4 & 7 \\ 6 & 9 \end{vmatrix}$$

M_{12} $\qquad$ M^T_{12}

$$|A^T| = a_{11} A_{11}^T + a_{12} A_{21}^T + \cdots + a_{1,k+1} A_{k+1,1}^T$$

$$= a_{11} A_{11} + a_{12} A_{12} + \cdots + a_{1,k+1} A_{1,k+1}$$

$$= |A|$$

Proved.

Properties of Determinants

Theorem 3

- If A has a row or column consisting entirely zeros, then $\det(A) = 0$.
- If A has two identical rows or columns, then $\det(A) = 0$.

Expansion according to the ith row

$$|A| = \sum_{j=1}^{n} a_{ij}A_{ij} = \sum_{j=1}^{n} a_{ij}(-1)^{i+j}\det(M_{ij})$$

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ a_{n1} & a_{n2} & a_{n3} & \dots & a_{nn} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \dots & a_{nn} \end{vmatrix}$$

Properties of Determinants

Lemma 1

Lemma: Let A be an $n \times n$ matrix. If A_{jk} denotes the cofactor of a_{jk} for $k = 1, \ldots, n$, then

$$a_{i1}A_{j1} + a_{i2}A_{j2} + \cdots + a_{in}A_{jn} = \begin{cases} \det(A) & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

Proof

- If $i = j$, then the left side is just the cofactor expansion of $\det(A)$ along the ith row.
- If $i \neq j$, let A^* be the matrix obtained by replacing the jth row of A by the ith row of A:

$$A^* = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \vdots & \vdots \\ a_{i1} & \cdots & a_{in} \\ \cdots & \cdots & \cdots \\ a_{i1} & \cdots & a_{in} \\ \vdots & \vdots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{bmatrix} \begin{matrix} \\ \\ j^{th} \text{ row} \\ \\ i^{th} \text{ row} \\ \\ \\ \end{matrix}$$

- By Theorem 3, $\det(A^*) = 0$
- Expand $\det(A^*)$ along j^{th} row:

$$\det(A^*) = a_{i1}A^*_{j1} + a_{i2}A^*_{j2} + \cdots + a_{in}A^*_{jn}$$
$$= a_{i1}A_{j1} + a_{i2}A_{j2} + \cdots + a_{in}A_{jn}$$
$$= 0$$

Eg. $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$ ← i-th, ← j-th

$i = 1$

$j = 2$

$$a_{11}A_{21} + a_{12}A_{22} + a_{13}A_{23} = 0$$

Define $A^* = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \\ 7 & 8 & 9 \end{bmatrix}$ ⟸ repeating the i-th row of A onto the j-th row

$|A^*| = 0$ since $R_i = R_j$

$$|A^*| = a^*_{21}A^*_{21} + a^*_{22}A^*_{22} + a^*_{23}A^*_{23}$$

along the j-th row

$$= a_{11}A^*_{21} + a_{12}A^*_{22} + a_{13}A^*_{23}$$

$$= a_{11}(-1)^{2+1}|M^*_{21}| + \cdots + a_{13}(-1)^{2+3}|M^*_{23}|$$

$$= a_{11}A_{21} + a_{12}A_{22} + a_{13}A_{23} = 0$$

$$A^*_{21} = (-1)^{2+1}\begin{vmatrix} 2 & 3 \\ 8 & 9 \end{vmatrix}$$

$$\| \quad \| \quad \| \quad \| \quad \|$$

$$A_{21} = (-1)^{2+1}\begin{vmatrix} 2 & 3 \\ 8 & 9 \end{vmatrix}$$

Properties of Determinants

Row Operation III : *A multiple of one row is added to another row.*

- Let E be the elementary matrix of type III formed from I by adding c times the ith row to the jth row, then

$$\det(EA) = \det(A)\,.$$

- In particular, E is a triangular matrix and all diagonal elements are 1, therefore $\det(E) = 1$, and

$$\det(EA) = \det(A) = \det(E)\det(A).$$

Proof

- If $\det(EA)$ is expanded by cofactors along j^{th} row:

$$\begin{aligned}\det(EA) &= (ca_{i1} + a_{j1})A_{j1} + \cdots + (ca_{in} + a_{jn})A_{jn} \\ &= (a_{j1}A_{j1} + \cdots + a_{jn}A_{jn}) + c(a_{i1}A_{j1} + \cdots + a_{in}A_{jn}) \quad \textit{(Lemma 1)} \\ &= (a_{j1}A_{j1} + \cdots + a_{jn}A_{jn}) = \det(A)\end{aligned}$$

E: $R_j + cR_i$

$$A=\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{i1} & a_{i2} & \cdots & a_{in} \\ a_{j1} & a_{j2} & \cdots & a_{jn} \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \xrightarrow{R_j+cR_i}$$

$\det(EA)=\det(A)$

$$\underbrace{EA}_{\| B}=\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{i1} & a_{i2} & \cdots & a_{in} \\ (ca_{i1}+a_{j1}) & (ca_{i2}+a_{j2}) & \cdots & ca_{in}+a_{jn} \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \leftarrow j\text{-th row}$$

$\det(EA)=\det(B)$

$=(ca_{i1}+a_{j1})B_{j1}+\cdots+(ca_{in}+a_{jn})B_{jn}$

$B_{j1}=A_{j1}$, $B_{jn}=A_{jn}$

$=c(a_{i1}A_{j1}+\cdots+a_{in}A_{jn})+(a_{j1}A_{j1}+\cdots+a_{jn}A_{jn})$

$a_{i1}A_{j1}+\cdots+a_{in}A_{jn}=0$

$a_{j1}A_{j1}+\cdots+a_{jn}A_{jn}$: cofactor expansion along the j-th row

$\det(A)$

$= \det(A)$

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(1) Type III.

$\det(EA) = \det(A)$.

(2) Eg. $I_2 \xrightarrow{R_2 - 2R_1} E = \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix}$

$I_2 \xrightarrow{R_1 + 3R_2} E = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$

Type III E is triangular, with diagonal entries being all 1s.

$\Rightarrow \det(E) = 1$

(3) $\det(EA) = \det(A) = 1 \cdot \det(A)$

$= \det(E)\det(A)$

Properties of Determinants

Row Operation I: *Two rows of A are interchanged*

- If A is an $n \times n$ matrix and E_{ij} is the $n \times n$ elementary matrix formed by interchanging the ith row and j^{th} row of I, then
$$\det(EA) = -\det(A).$$
- In particular, $\det(E_{ij}) = \det(E_{ij}I) = -\det(I) = -1$, then
$$\det(EA) = -\det(A) = \det(E)\det(A).$$

Proof (hint)

- By induction
- Expand the determinant along an unchanged row.

Type I: $R_i \leftrightarrow R_j$.

Mathematical Induction.

① $n=2$ $\quad |A| = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}a_{21}$

$$|EA| = \begin{vmatrix} a_{21} & a_{22} \\ a_{11} & a_{12} \end{vmatrix} = a_{12}a_{21} - a_{11}a_{22} = -|A|$$

② $n=3$ $\quad R_1 \leftrightarrow R_2$

$$|\underbrace{EA}_{=A^*}| = \begin{vmatrix} a_{21} & a_{22} & a_{23} \\ a_{11} & a_{12} & a_{13} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

$$= a_{31}\underbrace{A^*_{31}}_{(-A_{31})} + a_{32}\underbrace{A^*_{32}}_{(-A_{32})} + a_{33}\underbrace{A^*_{33}}_{(-A_{33})}$$

$$= -(a_{31}A_{31} + a_{32}A_{32} + a_{33}A_{33}) = -\det(A) \quad -A_{3j}$$

$$A^*_{3j} = (-1)^{3+j} |M^*_{3j}| = (-1)^{3+j}(-1)|M_{3j}|$$

$$A_{3j} = (-1)^{3+j} |M_{3j}|$$

$M_{3j} \xrightarrow{R_1 \leftrightarrow R_2} M^*_{3j}$

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

(3) Assume when $n=k$. Type I.

$\det(EA) = -\det(A)$

(4) When $n=k+1$.

$R_i \leftrightarrow R_j$

A^*

$|EA| = a_{m1} A^*_{m1} + \cdots + a_{mn} A^*_{m,k+1}$

$A^*_{m1} = (-A_{m1})$

$A^*_{m,k+1} = (-A_{m,k+1})$

cofactor expansion along the row that is not interchanged.

$m \neq i$ & $m \neq j$.

$A_{m1} = (-1)^{m+1} |M_{m1}|$ ($k \times k$) Type I

$A^*_{m1} = (-1)^{m+1} |M^*_{m1}|$ ($k \times k$)

$= (-1)^{m+1} (-1) |M_{m1}|$

$|EA| = -(a_{m1} A_{m1} + \cdots + a_{m,k+1} A_{m,k+1})$

$= -|A|$

(1) Type I.

$\det(EA) = -\det(A)$. $\forall$ $n \times n$ A.

(2) If $A = I_n$

$\det(EI) = -\det(I) = -1$

$\det(EI) = \det(E)$

(3) $\det(EA) = -\det(A) = (-1)\det(A) = \det(E)\det(A)$

Properties of Determinants

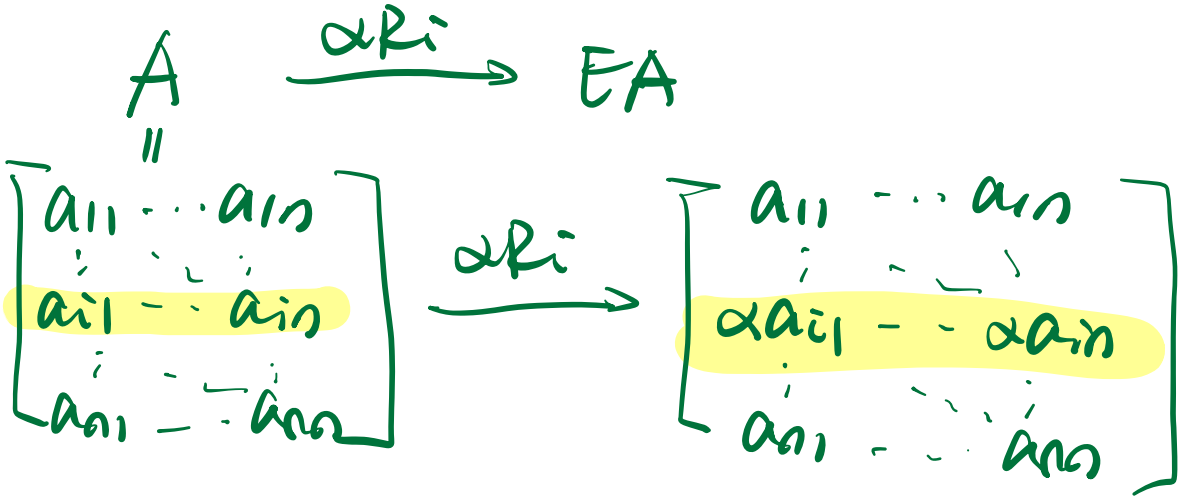
Row Operation II : A *row of A is multiplied by a nonzero constant.*

- Let E be the elementary matrix of type formed by multiplying the ith row by a nonzero constant α, then

$\alpha \in \mathbb{R}$, $\alpha \neq 0$ $\det(EA) = \alpha \det(A)$.

- In particular, $\det(E) = \det(EI) = \alpha\det(I) = \alpha$, then

$$\det(EA) = \alpha\det(A) = \det(E)\det(A).$$



$$|EA| = (\alpha a_{i1})A^*_{i1} + \cdots + (\alpha a_{in})A^*_{in}$$

($|EA| = |A^*|$; $A^*_{i1} = A_{i1}$, $A^*_{in} = A_{in}$)

$$= \alpha(a_{i1}A_{i1} + \cdots + a_{in}A_{in})$$

$$= \alpha|A|$$

(1) Type II.

$$\det(EA) = \alpha \det(A) \qquad \forall\ n\times n\ A.$$

(2) If $A = I$

$$\det(EI) = \alpha \det(I)$$

$$\det(E) = \alpha \cdot 1 = \alpha, \quad \alpha \neq 0$$

(3) $\det(EA) = \alpha \cdot \det(A) = \det(E)\det(A)$.

Determinant and elementary matrices

In summary, if E is an elementary matrix, then

$$\det(EA) = \det(E)\det(A)$$

$\neq 0$

where

$$\det(E) = \begin{cases} -1 & \text{if } E \text{ is of type I} \\ \alpha \neq 0 & \text{if } E \text{ is of type II} \\ 1 & \text{if } E \text{ is of type III} \end{cases}$$

Similar results hold for column operations:

$$\det(AE) = \det(A)\det(E)$$

$\det(A) = \det(A^T)$

$ab = ba$ ✓ $AB \neq BA$

because if E is an elementary, then E^T is also an elementary matrix and

$$\det(AE) = \det\left((AE)^T\right) = \det\left(E^T A^T\right) = \det\left(E^T\right)\det\left(A^T\right)$$

$$= \det(E)\det(A) = \det(A)\det(E)$$

Determinant and inevitability

Theorem

An $n \times n$ matrix A is singular if and only if $\det(A) = 0$.

Proof

- Matrix A can be reduced to row echelon form with a finite number of row operations. Thus,

$$U = E_k E_{k-1} \dots E_1 \mathrm{A}$$

- Then,

$$\det(U) = \det(E_k)\det(E_{k-1}) \dots \det(E_1)\det(A)$$

- Therefore, $\det(A) = 0$ if and only if $\det(U) = 0$. If A is nonsingular, then U is triangular with all diagonal equals to 1, $\det(U) = 1$. If A is singular, U contains a row of all zeros, $\det(U) = 0$.

Motivation.

A is nonsingular $\iff$ $\det(A) \neq 0$

A is singular $\iff$ $\det(A) = 0$

A: $n \times n$.

A is nonsingular $\iff$ A is row equivalent to I

A is singular $\iff$ The r.r.e.f of A has a zero row

Assume U is the r.r.e.f of A.

$U = E_k \cdots E_2 E_1 A$

$\det(EA) = \det(E)\det(A)$

$\det(U) = [\det(E_k) \cdots \det(E_2)\det(E_1)]\det(A)$

$\det(E_k) \neq 0$, $\det(E_2) \neq 0$, $\det(E_1) \neq 0$

$\alpha \neq 0$

$\det(U) = \alpha \det(A)$

$\alpha \neq 0$

$\Downarrow$

$\det(A) = 0 \iff \det(U) = 0$

$\det(A) \neq 0 \iff \det(U) \neq 0$

(1) A is nonsingular $\iff$ A row equivalent to I $\iff$ $U = I$ $\iff$ $\det(U) \neq 0$

$\iff \det(A) \neq 0$

(2) A is singular $\iff$ A is not row equivalent to I $\iff$ There is a zero row in U (r.e.f) $\iff$ $\det(U) = 0$

$\iff \det(A) = 0$

Properties of Determinants

Theorem 4

If A and B are $n \times n$ matrices, then

$\det(AB) = \det(A)\det(B)$.

Proof

- Suppose B is singular matrix, then $B\boldsymbol{x} = 0$, has nontrivial solutions, then $\boldsymbol{x}$ is also a nontrivial solution to $AB\boldsymbol{x} = 0$. Then AB is also a singular matrix. So $\det(AB) = 0 = \det(A) \cdot \det(B) = \det(A) \cdot 0 = 0$.
- If B is not singular, then is row equivalent to the identity matrices or $B = E_1E_2 \cdots E_k$ for some k. And $\det B = \det(\mathrm{E}_1)\det(E_2) \cdot \det(E_k)$. (This is due to the property of elementary matrices)
- $\det(AB) = \det(AE_1E_2 \cdots E_k) = \det(A)\det(E_1) \cdot \ldots \cdot \det(E_k) = \det(A) \cdot \det(B)$

$\det(AB) =$ _____

$m \times n$ $n \times n$

$\det(AB)_{m \times m}$ exists $\neq$ $\det(A)_{m \times n}$ $\det(B)_{n \times n}$ does not exist

B: $n \times n$ $\det(B) \begin{cases} \neq 0 \\ = 0 \end{cases}$

① $\det(B) = 0$

$\Downarrow$

B is singular. $\Leftrightarrow$ $Bx = 0$ has a non-trivial solution. $x_0 \neq 0$.

$\Downarrow$

$(AB)x_0 = A0 = 0$, $x_0 \neq 0$

$\Downarrow$

$(AB)x = 0$ has a non-trivial solution $x_0 \neq 0$

$\Downarrow$

AB is singular

$\Updownarrow$

$\det(AB) = 0 = \det(A)0 = \det(A)\det(B)$

(2) Case 2. $\det(B) \neq 0$

$\Updownarrow$

B is nonsingular.

$\Updownarrow$

B is row equivalent to I.

$\Updownarrow$

$B = E_k E_{k-1} \cdots E_2 E_1 I$

$= E_k E_{k-1} \cdots E_2 E_1$

$\Downarrow$

$\det(AB) = \det(A E_k E_{k-1} \cdots E_1)$

$= \det(A) \det(E_k) \cdots \det(E_1) \det(I)$

$= \det(A) \det(E_k \cdots E_1 I)$

B

$= \det(A) \det(B)$.

$\det(EA) = \det(E)\det(A)$.

$\det(AE) = \det(A)\det(E)$.

(1) & (2) $\Rightarrow$ $\det(AB) = \det(A)\det(B)$

$n \times n$ A, B $\quad$ $n \times n$ $\quad$ $n \times n$

$AB \neq BA$ $\quad$ $\det(AB) = \det(A)\det(B) = \det(B)\det(A)$

$= \det(BA)$.

Exercise

For a square matrix A, what can you conclude about $\det(AA^T)$?

$\det(A)\det(A^T) = \det(A)\det(A) = \det(A)^2$

Exercise

For a square matrix A, what can you conclude about $\det(AA^T)$?

Solution

$$\det(AA^T) = (\det(A))^2$$

Properties of Determinants

proofs are required.

1. If A is nonsingular, then det(A)≠0 and $\det(A^{-1}) = \frac{1}{\det(A)}$
2. $\det(kA) = k^n \det(A)$ (n×n)

Proof

$$|A| \cdot |A^{-1}| = |A \cdot A^{-1}| = |I| = 1 \quad \Rightarrow \quad |A^{-1}| = \frac{1}{|A|}$$

$|A| \neq 0$

$A \xrightarrow{kR_1} \xrightarrow{kR_2} \cdots \xrightarrow{kR_n} kA$

$A \xrightarrow{kR1, kR2, \ldots, kRn} kA \Rightarrow |kA| = k^n |A|$

$E_1 \quad E_2 \quad E_n$

$$2\begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 2 \\ 4 & 4 \end{bmatrix}$$

$kA = E_n E_{n-1} \cdots E_1 A$

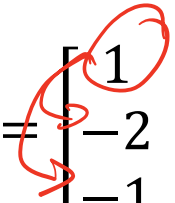
$\det(kA) = \det(E_n) \cdots \det(E_1) \det(A)$

$k \quad \cdots \quad k \quad = k^n \det(A)$

$$\begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix} \xrightarrow{2R_1} \xrightarrow{2R_2} \begin{bmatrix} 2 & 2 \\ 4 & 4 \end{bmatrix}$$

Compute determinant

- **Example 1** Compute det *A*, where $A = \begin{bmatrix} 1 & -4 & 2 \\ -2 & 8 & -9 \\ -1 & 7 & 0 \end{bmatrix}$



Compute determinant

- **Example 1** Compute det *A*, where $A = \begin{bmatrix} 1 & -4 & 2 \\ -2 & 8 & -9 \\ -1 & 7 & 0 \end{bmatrix}$
- **Solution** The strategy is to reduce *A* to echelon form and then to use the fact that the determinant of a triangular matrix is the product of the diagonal entries. Type III row operations do not change the determinant:

$$|A| = \begin{vmatrix} 1 & -4 & 2 \\ -2 & 8 & -9 \\ -1 & 7 & 0 \end{vmatrix} = \begin{vmatrix} 1 & -4 & 2 \\ 0 & 0 & -5 \\ -1 & 7 & 0 \end{vmatrix} = \begin{vmatrix} 1 & -4 & 2 \\ 0 & 0 & -5 \\ 0 & 3 & 2 \end{vmatrix}$$

$R2 + 2R1$ $R3 + R1$

Compute determinant

- An interchange of rows 2 and 3 reverses the sign of the determinant, so

$$detA = -\begin{vmatrix} 1 & -4 & 2 \\ 0 & 3 & 2 \\ 0 & 0 & -5 \end{vmatrix} = -(1)(3)(-5) = 15$$

Eg.

$$\begin{vmatrix} 100 & 200 & 300 \\ 1 & 3 & 5 \\ 2 & 4 & 5 \end{vmatrix} = ? \begin{vmatrix} 1 & 2 & 3 \\ 1 & 3 & 5 \\ 2 & 4 & 5 \end{vmatrix}$$

A

100 ✓

$\frac{1}{100}$ ✗

B

$A \xrightarrow{\text{Too } R_1} B = EA$

$\det(B) = \det(E)\det(A)$

$\det(A) = \frac{1}{\det(E)} \det(B)$

Example

Evaluate the determinant of $\begin{vmatrix} 3 & -1 & 2 \\ 4 & -6 & 9 \\ -2 & 3 & -3 \end{vmatrix} = |A|$

$$|A| = \begin{vmatrix} 3 & -1 & 2 \\ 0 & 0 & 3 \\ -2 & 3 & -3 \end{vmatrix} = 3 \times (-1)^{2+3} \begin{vmatrix} 3 & -1 \\ -2 & 3 \end{vmatrix}$$

$R_2 + 2R_3$

$$= (-3) \times (9 - 2) = -21$$

Example

Evaluate the determinant of $\begin{vmatrix} 3 & -1 & 2 \\ 4 & -6 & 9 \\ -2 & 3 & -3 \end{vmatrix}$

Solution

$$\begin{vmatrix} 3 & -1 & 2 \\ 4 & -6 & 9 \\ -2 & 3 & -3 \end{vmatrix} \underset{R2+2R3}{=} \begin{vmatrix} 3 & -1 & 2 \\ 0 & 0 & 3 \\ -2 & 3 & -3 \end{vmatrix} = (-3)\begin{vmatrix} 3 & -1 \\ -2 & 3 \end{vmatrix} = -21$$

A simpler way to find the determinant.

Example

- Find |A| for $A = \begin{bmatrix} -2 & 3 & 4 \\ 3 & 3 & 5 \\ 4 & -6 & -8 \end{bmatrix}$

$|A| = \begin{vmatrix} -2 & 3 & 4 \\ 3 & 3 & 5 \\ 0 & 0 & 0 \end{vmatrix} = 0$

$R_3 + 2R_1$

Example

If A is a 2×2 matrix with $|A| = 4$, use Theorems to compute the following determinants.

$(a)|3A|$ $(b)|A^2|$ $(c)|5A^T A^{-1}|$, assuming A^{-1} exists

$= 3^2|A|$

$= 9 \times 4$

$= 36$

$= |A \cdot A|$

$= |A||A|$

$= |A|^2$

$= 16$

$= 5^2|A^T A^{-1}|$

$= 25|A^T||A^{-1}|$

$= 25|A|\frac{1}{|A|}$

$= 25$

Example

If A is a 2×2 matrix with $|A| = 4$, use Theorems to compute the following determinants.

$(a)|3A|$ $(b)|A^2|$ $(c)\left|5A^T A^{-1}\right|$, assuming A^{-1} exists

Solution

(a) $|3A| = (3^2)|A| = 9 \times 4 = 36.$

(b) $|A^2| = |AA| = |A|\,|A| = 4 \times 4 = 16.$

(c) $|5A^T A^{-1}| = (5^2)|A^T A^{-1}| = 25|A^T||A^{-1}| = 25$

Exercise

For $C = \begin{bmatrix} k & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & k \end{bmatrix}$, find all values of $k \in R$, such that $Cx = 0$ has a nontrivial solution. $\Leftrightarrow$ C is singular $\Leftrightarrow$ $\det(C) = 0$

$$\det(C) = \begin{vmatrix} k & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & k \end{vmatrix}$$

$R_2 - R_3$
$R_1 - kR_3$

$$\begin{vmatrix} 0 & 1-k & 1-k^2 \\ 0 & 1 & 1-k \\ 1 & 1 & k \end{vmatrix} = 1 \times (-1)^{3+1} \begin{vmatrix} 1-k & 1-k^2 \\ 1 & 1-k \end{vmatrix}$$

$$= (1-k)^2 - (1-k^2)$$

$= 1 + k^2 - 2k + 1 + k^2$

$= 2k^2 - 2k = 0$

$\Rightarrow k(k-1) = 0$

$\Rightarrow k = 0$ or $k = 1$

Exercise

For $C = \begin{bmatrix} k & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & k \end{bmatrix}$, find all values of $k \in R$, such that $Cx = 0$ has a nontrivial solution.

Solution

$k = 0 \; or \; k = 1$

Exercise

For $D = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, find an equation that λ satisfies so that $(D - \lambda I)x = 0$ has a nontrivial solution.

$(D - \lambda I)x = 0$ has a nontrivial solution.

$\Leftrightarrow$ "A"

$\Leftrightarrow$ $\det(D - \lambda I) = 0$

$$\det(D - \lambda I) = \begin{vmatrix} a-\lambda & b \\ c & d-\lambda \end{vmatrix} = 0$$

$\Leftrightarrow$

$$(a-\lambda)(d-\lambda) - bc = 0$$

$$\Rightarrow \lambda^2 - (a+d)\lambda + ad - bc = 0$$

Exercise

For $D = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, find an equation that λ satisfies so that $(D - \lambda I)x = 0$ has a nontrivial solution.

Solution

$$\lambda^2 - (a + d)\lambda + ad - bc = 0$$

Exercise

Let x and y be vectors in R^n with $n > 1$. Prove that if $A = \mathbf{x}\mathbf{y}^T$ then $\det(A) = 0$.

Proof

$$A = \begin{bmatrix} a_1b_1 & a_1b_2 & \dots & a_1b_n \\ a_2b_1 & a_2b_2 & \dots & a_2b_n \\ \vdots & \vdots & \vdots & \vdots \\ a_nb_1 & a_nb_2 & \dots & a_nb_n \end{bmatrix}$$

Perform row operations of type II.

$$|A| = a_1 \begin{vmatrix} b_1 & b_2 & \dots & b_n \\ a_2b_1 & a_2b_2 & \dots & a_2b_n \\ \vdots & \vdots & \vdots & \vdots \\ a_nb_1 & a_nb_2 & \dots & a_nb_n \end{vmatrix} = a_1a_2 \begin{vmatrix} b_1 & b_2 & \dots & b_n \\ b_1 & b_2 & \dots & b_n \\ \vdots & \vdots & \vdots & \vdots \\ a_nb_1 & a_nb_2 & \dots & a_nb_n \end{vmatrix} = 0$$

Because the first two rows are identical.

$$X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}_{n\times 1} \qquad y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}_{n\times 1}$$

$$A = \underset{n\times n}{x}\ \underset{n\times 1}{y^T} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}_{n\times 1} \begin{bmatrix} y_1 & y_2 & \cdots & y_n \end{bmatrix}_{1\times n}$$

$$= \begin{bmatrix} x_1y_1 & x_1y_2 & \cdots & x_1y_n \\ x_2y_1 & x_2y_2 & \cdots & x_2y_n \\ \vdots & \vdots & \ddots & \vdots \\ x_ny_1 & x_ny_2 & \cdots & x_ny_n \end{bmatrix}_{n\times n}$$

$$|A| = x_1 \begin{vmatrix} y_1 & y_2 & \cdots & y_n \\ x_2y_1 & x_2y_2 & \cdots & x_2y_n \\ \vdots & & & \\ x_ny_1 & - & - & x_ny_n \end{vmatrix} = x_1x_2 \begin{vmatrix} y_1 & y_2 & \cdots & y_n \\ y_1 & y_2 & \cdots & y_n \\ & & & \\ x_ny_1 & - & - & x_ny_n \end{vmatrix}$$

$-$

True or False

$A: n \times n$ $B: n \times n$

T
1. $\det(AB) = \det(BA)$

F
2. $\det(A + B) = \det(A) + \det(B)$

F
3. $\det(cA) = c \det(A)$ $= c^n \det(A)$

T
4. $\det((AB)^T) = \det(A) \det(B)$

F
- $\det(A) = \det(B) \Rightarrow A = B$ $\rightarrow A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ $B = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{bmatrix}$

T
- $\det(A^k) = \det(A)^k$ $\det(A^k) = \det(AA \cdots A) = \det(A) \cdots \det(A)$

T
- A triangular matrix is nonsingular if and only if its diagonal entries are all nonzero. $\det(A) = a_{11} a_{22} \cdots a_{nn} \neq 0$ $\det(A)^k$

$\Leftrightarrow$ $a_{ii} \neq 0$ $i = 1, \ldots, n$

(2) $\det(I_2 + I_2) = \begin{vmatrix} 2 & 0 \\ 0 & 2 \end{vmatrix} = 4 \neq$

$\det(I_2) + \det(I_2) = 1 + 1 = 2$

(4) $\det((AB)^T) = \det(AB) = \det(A) \det(B)$

$\det(AB) = \det(A) \det(B) = \det(B) \det(A)$